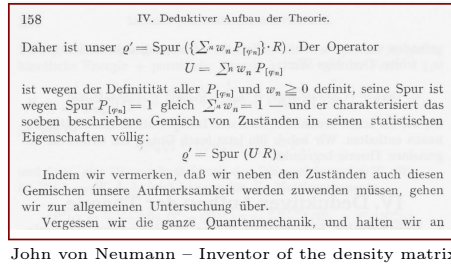


PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



EXERCISE 10.1: DENSITY MATRIX OF A SINGLE QUBIT

(4P)

For a given ensemble of qubits the observables

$$\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 0 & 2i \\ -2i & 0 \end{pmatrix}$$

have been measured many times, giving the expectation values

$$\langle \mathbf{A} \rangle = 2, \quad \langle \mathbf{B} \rangle = 1/2, \quad \langle \mathbf{C} \rangle = 0.$$

- Determine the density matrix of the ensemble.
- Is the ensemble pure or mixed?
- If we measure σ^z , what is the probability to get +1 ?
- Determine the expectation values $\langle \sigma^x \rangle, \langle \sigma^y \rangle, \langle \sigma^z \rangle$.

EXERCISE 10.2: CRITERION FOR THE EQUIVALENCE OF ENSEMBLES

(8P)

Consider a statistical ensembles $\{|\psi_i\rangle, p_i\}$ with the associated density matrix

$$\rho = \sum_{i=1}^N p_i |\psi_i\rangle \langle \psi_i|.$$

Here N is arbitrary (smaller or larger than the dimension of the Hilbert space) and the $|\psi_i\rangle$ are pairwise different and normalized but not necessarily orthogonal. Moreover, let

$$\rho = \sum_{k=1}^K \lambda_k |k\rangle \langle k|$$

be the spectral decomposition of ρ with orthonormal eigenvectors $|k\rangle$ and eigenvalues λ_k . The aim of this exercise is to work out an equivalence theorem for different ensembles with the same density matrix.

- Show that every vector $|\psi_i\rangle$ can be written as a linear combination of the eigenvectors $|k\rangle$, that is, prove that $|\psi_i\rangle \in \text{span}(\{|k\rangle\})$. (2P)

- (b) Use (a) to prove that every $|\psi_i\rangle$ can be written as

$$\sqrt{p_i}|\psi_i\rangle = \sum_{k=1}^K c_{ik} \sqrt{\lambda_k} |k\rangle$$

with coefficients c_{ik} obeying $\sum_{i=1}^N c_{ik} c_{il}^* = \delta_{k,l}$ for all $k, l = 1, \dots, K$. (2P)

- (c) Because of (a) we know that $N \geq K$. For $N = K$ the result obtained in (b) would imply that the coefficient matrix c_{ik} is unitary. Show that for $N > K$ the rectangular matrix c_{ik} can be extended to a quadratic unitary matrix in such a way that the relation proven in (b) still holds. (1P)
- (d) So far we have proven that if $\sum_{i=1}^N p_i |\psi_i\rangle \langle \psi_i| = \sum_{k=1}^K \lambda_k |k\rangle \langle k|$, then there exists a unitary $N \times N$ -matrix c_{ik} such that $\sqrt{p_i}|\psi_i\rangle = \sum_{k=1}^K c_{ik} \sqrt{\lambda_k} |k\rangle$. Prove the statement in opposite direction. (1P)
- (e) Use (a)-(d) to show that $p_i = \sum_{k=1}^K T_{ik} \lambda_k$ (or short: $\vec{p} = \mathbf{T} \vec{\lambda}$), where $\mathbf{T}$ is a so-called *doubly-stochastic matrix*, that is, all matrix entries are real and non-negative and the sums over rows and columns give 1, i.e. $\sum_{i=1}^N T_{ik} = \sum_{k=1}^K T_{ik} = 1$. (1P)
- (f) Use e.g. *Mathematica*[®] to work out the following explicit example with $N = K = 2$

$$p_1 = p_2 = \frac{1}{2}, \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and determine $\lambda_{1,2}$, $|\phi_{1,2}\rangle$, c_{ij} , and T_{ij} . Check the properties derived in (e). (1P)

($\Sigma = 12\mathbf{P}$)

Please submit your solution electronically as a single PDF file until Wednesday, June 24, 2026, via WueCampus.